Indian Institute of Technology (Banaras Hindu University), Varanasi, U.P.

Mid Semester Examination, Sept' 2025

Duration:	: 02 Hours	Maximum marks	: 30
Programme	: M.Tech. Mining Engg. (Pt-I) and Ph.D	Semester	: Odd
Course Title	: Reliability Engineering	Course Code	: MN572

Note:

1) *Answer all questions.*
2) *Marks are indicated against each question in the brackets.*
3) *Take* $Z_{0.05} = \mp 1.645$; $Z_{0.12} = \mp 1.175$ and $t_{0.05,9} = 1.833$ *where applicable*

Q. 1 (a) Define and explain the terms 'Reliability' and 'Maintainability'. [2]

(b) Discuss mode (s) and cause(s) of failures with suitable examples. [2]

(c) What are the various methods for improving system reliability? [1]

(d) What are 'probability mass function' and 'probability density function'? [1]

(e) If $T \geq 0$ is a continuous random variable that represents the time to failure of a system, then give the mathematical expressions for the reliability function $R(t)$, cumulative distribution function $F(t)$, probability density function $f(t)$ and failure rate function $\lambda(t)$ at time t. Establish a relationship among these functions. [4]

Q. 2 (a) The Company 'X' manufactures chipset for use on smartphone. The time to failure, in year, of this chipset has the following PDF (Probability Density Function) [5]

$$f(t) = \frac{200}{(t+10)^3} \qquad for\ t \geq 0$$

(i) Determine the reliability of the chipset after first year of operation.
(ii) Compute the MTTF.
(iii) What is the design life for a reliability of 0.95?
(iv) Will a one-year burn-in period improve the reliability in part (i)? If so, what is the new reliability?

(b) A household appliance is advertised as having more than a 10-yr life. The PDF of the appliance is given by [3]

$$f(t) = 0.1(0.05t + 1)^{-3} \quad for\ t\ (in\ years) \geq 0$$

If it has survived a 1-year warranty period, what is its MTTF after the warranty period?

(c) The time to failure of an electric bulb follows a Weibull distribution with scale parameter of 2000 hours and shape parameter of 0.5. What is the probability that the electric bulb will last more than 4000 hours? What is the mean time to failure? [2]

Continued.....

Q.3 (a) The failure of a critical component of mining dozer is normally distributed with 90% of the failure occurring between 200 hours and 270 hours of operation, meaning 5% failure occurs before 200 hours of operation and 5% failure occurs after 270 hours of operation. [5]

(i) Calculate MTTF and σ of failure times of this component.
(ii) What is the reliability at the time of replacement when the component is replaced after 210 hr. of use?
(iii) Determine the design life if not more than a 1% probability of failure prior to the replacement is tolerated.
(iv) Estimate the reliability for another 10 hours of use if the component has been operating for 200 hours.

(b) Given the following 10 failures times in hours of an item X; [5]

24.5, 18.9, 54.7, 48.2, 20.1, 29.3, 15.4, 33.9, 72.0, 86.1.

Estimate $R(t)$, $F(t)$, $f(t)$ and $\lambda(t)$ of the component X at different time intervals for this ungrouped failure data set and compute a 90% confidence interval for the MTTF of the item X.

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